

Online Hierarchical IRT with Assumed-Density Filtering for Predictive AI Evaluation

Haoyuan Luo — University of California, Davis — zixluo@ucdavis.edu

Technical report — *Predictive AI Evaluation Competition, NeurIPS 2026*. Codabench submission **955485** (paiec_irt.zip, sha256 98555787...2cfe0f). Code: <https://github.com/lucaluo925/paiec-2026>

Abstract

A decision maker choosing between AI systems can rarely afford to evaluate them on every item. This competition asks how well a system’s per-item success can be predicted on a benchmark never seen during development, given a budget of 0 to 31 labels for the subject-benchmark pair under test. We model the response as a hierarchical item-response model — $\text{logit } P(\text{correct}) = a - \theta_s + v_s - \beta - d_i - \sum w_f$ — with subject ability θ_s calibrated offline on the public measurement-db and predicted from provider, release date, parameter count and name tokens for subjects absent from calibration. At evaluation time we hold one joint Gaussian posterior per anonymous benchmark and absorb each acquired label with an assumed-density-filtering step (probit approximation, closed-form rank-one update). Labels of *other* subjects on the same benchmark inform the shared latents, which is where most of the gain at small budgets comes from.

Under leave-one-benchmark-out validation on four public benchmarks (147 subject-benchmark pairs, 19,621 evaluation targets), the fully out-of-fold model reaches Brier ALC **0.16875** against **0.23110** for the organizers’ empirical-mean baseline and 0.25000 for a constant 0.5; cross-fitting the shrinkage weights under the same protocol gives an honest **0.16515**. The advantage does not depend on having met the model before: on the 27 pairs whose identity calibration never saw, ALC is 0.18424 against 0.24960.

Two negative results are reported as fully as the positive one. First, the competition’s formative feedback is far noisier than it appears: five byte-identical submissions of the same artefact scored 0.20624, 0.20098, 0.18124, 0.18685 and 0.19461 — a standard deviation of **0.0102** on ALC and 0.014–0.023 per budget, because each evaluation redraws only eight or nine subject-benchmark pairs. Of everything that feedback could be asked, only three statements survive it, and we state which. Second, three pre-registered mechanism-level fixes for the zero-label prior — the one budget where the live result clearly exceeds our local estimate — were all falsified on leave-one-benchmark-out, and the shipped configuration was left unchanged. We also record a sequencing error of our own: the noise calibration was run after, not before, the first live score had been used to draw conclusions.

Keywords: predictive evaluation; item response theory; assumed density filtering; online Bayesian updating; calibration; label-efficient evaluation; Brier score.

Scope of the numbers. Every figure outside §5b is a local hold-out estimate on the public measurement-db, produced by the scripts in this repository. §5b reports the live Codabench scores and treats them as a repeated measurement of noise, not as a selection signal; the leaderboard was never consulted for model selection.

1. Task and scoring

A predictor receives a subject (eight visible attributes: `normalized_name`, `provider`, `release_date`, `access_date`, `harness`, `reasoning_effort`, `harness_version`, `subject_features_extra`), an item (`item_content`, `item_features`, `interactors`, `anonymous_benchmark_id`) and the labels acquired so far, and outputs $P(\text{correct})$. Brier scores at label budgets $b \in \{0, 1, 3, 7, 15,$

31} are combined as $ALC = 0.1 \cdot B_0 + 0.2 \cdot B_1 + 0.2 \cdot B_3 + 0.2 \cdot B_7 + 0.2 \cdot B_{15} + 0.1 \cdot B_{31}$, so 80 % of the weight lies at 1–15 labels. The official rules average Brier within each subject–benchmark pair first and then across pairs with equal weight, “regardless of its number of responses”; all numbers in this report use that pair-macro aggregation. Test benchmarks are private and unseen during development.

2. Data

- **Training corpus.** The public aims-foundations/measurement-db (Hugging Face, dataset revision bc8204d811823da849c6686bf124d4ca9f82e4de), restricted — like the organizers’ preparation script — to benchmarks with `response_type = binary` and `granularity = item`.
- **What that leaves.** The repository holds 10 benchmark directories (`response_type`: 6 binary, 2 mixed, 2 fraction). Two of the six “binary” directories are reproduction_checks/.../tables smoke artifacts carrying 1 and 2 responses and re-using their parent’s `benchmark_id`; they are not independent benchmarks and are excluded (they would also fail the ≥ 80 -item rule). **Four usable benchmarks remain**, and neither of the two excluded top-level benchmarks (matharena, mmdocrag) is replaced, so the training pool contains no mathematics and no document-understanding benchmark.

benchmark	domain	subjects	items	responses	density	base rate
multi_swebench	software engineering, agents	82	2 126	57 808	0.332	0.148
researchcodebench	software engineering, ML	31	212	6 572	1.000	0.352
swe_rebench	software engineering, agents	1	6 306	6 306	1.000	0.475
real_webagents	agents and tool use	33	233	3 759	0.489	0.375

In total 74 445 responses, 8 877 items, 147 subject configurations and 60 distinct model identities. 17 identities appear in ≥ 2 benchmarks and **none in ≥ 5** , so the cross-benchmark ability calibration rests on 17 models.

- **Attribute coverage.** `normalized_name` and `provider` 100 %, `release_date` 93.9 %, `access_date` and `harness` 55.8 %, `harness_version` 0.7 %, **reasoning_effort** and **subject_features_extra** 0 %. The configuration-effect terms for reasoning effort therefore have no support in this corpus. `item_features` is non-empty for 2 571 items (most common keys `lang`, `website`, `paper`); `interactors` is empty everywhere.
- **Preprocessing.** One response per (subject, item, interactors): the lowest (test condition, trial) row with a 0/1 grade, because the visible input cannot distinguish trials or conditions. Missing attributes become "".
- **No other data.** No external data, benchmark results, model cards or web sources were used. The model registry used for normalized names is the one already embedded in measurement-db subject tables.

3. Local evaluation protocol

All model selection used a local replica of the evaluator (`eval/local_scorer.py`), never the Codabench leaderboard.

- **Streaming protocol.** Re-implements `streaming_alc_v1` from the organizers' `tools/streaming_ingestion.py`: benchmarks with ≥ 80 items; a 50/50 acquisition/evaluation item split **per subject-benchmark pair**; each pair's acquisition items are streamed one at a time with `max_labels = 31`; the default policy queries when a SHA-256 uniform falls below `labels_remaining / items_remaining`; all pairs reach budget `b` before the checkpoint at `b`; `labeled` contains the labels of all sampled pairs.
- **Checked against the official rules page** (aimslab.stanford.edu/competition): the per-pair 50/50 split, the pair-macro averaging, the ALC weights, the budgets, `max_labels = 31`, the context fields (`subject_id`, `benchmark_id`, `labels_acquired`, `labels_remaining`, `max_labels`, `items_remaining` including the current candidate), and the global scope of `labeled` all match. Two of our earlier assumptions did **not** match and were corrected: the split had been made once per benchmark and shared across subjects, and Brier had been averaged over predictions rather than over pairs. Both corrections are locked by regression tests (`test_official_protocol_defaults`, `test_pair_macro_weights_pairs_equally`).
- **Verification.** (i) A constant 0.5 predictor scores exactly 0.25 at every budget and in ALC (weights are integers in tenths and summed with `math.fsum`).
(ii) The organizers' own streaming client (`run_streaming`) driven by our coordinator reproduces our acquisitions and predictions exactly (unit test).
(iii) Label sets are nested across budgets.
- **Hold-out.** Two-way: benchmarks are split into 5 folds (benchmarks sharing $\geq 5\%$ item texts are kept together), and 20 % of model identities are removed from every training fold and appear only as unseen subjects. Items are never split at random. With four benchmarks the 5-fold split degenerates to **leave-one-benchmark-out** (fold sizes 1, 1, 1, 1, 0). Local evaluation uses every subject configuration in the corpus: 147 pairs and 19 621 evaluation targets, of which 27 pairs / 3 556 targets belong to the 12 held-out identities.
- **Still unverified.** The rules page does not state the evaluation worker concurrency or the size of the private test set. The engine holds an `RLock` and is safe under threads; process-level parallelism only re-creates state. The caps we pass locally (≤ 200 subjects per benchmark, ≤ 200 evaluation items per pair, ≤ 800 streamed items per pair) are ours, not the rules'; at these values none of them binds on this corpus, so the reported numbers use the whole public dataset. The rules' own cap — 1 000 unique subject-item pairs — applies to formative feedback runs only.

4. Method

4.1 Model

For benchmark B , subject s and item i ,

$$\text{logit } P(y=1) = a \cdot \theta_s + v_s - \beta - d_i - \sum_f w_f$$

- θ_s — general ability, **calibrated offline**. Known models use their fitted ability; unseen models use a ridge regression on provider, release date, parameter count (parsed from the name) and name tokens, plus learned effects of reasoning effort / harness settings.
- a, β — the benchmark's discrimination of general ability and its difficulty; priors are the across-benchmark distribution of item-level fits.
- $v_s \sim N(0, \tau^2 + (\bar{a}^2 + \text{var } a) \cdot \text{var } \theta_s)$ — subject $\times$ benchmark deviation.
- $d_i \sim N(0, \sigma_d^2)$ — item difficulty; $w_f \sim N(0, \sigma_w^2)$ — effects of the item's `item_features` / interactors tokens, learned within the benchmark.

4.2 Offline calibration (train/fit_offline.py)

1. Subject abilities by weighted alternating least squares on per-(benchmark, subject) logit accuracies: $\text{logit}(\text{acc}) = a_B \theta_u - b_B$, with θ of a configuration unit shrunk towards its model's ability plus configuration effects.
2. Item side: for every training benchmark, an item-level model (a, β, v_s, d_i with variance components τ^2, σ_δ^2) by block-coordinate Newton with EM variance updates. The distribution of these item-side parameters across benchmarks gives the priors for a new benchmark. Item-level parameters are never transferred directly, because the test benchmarks are new.
3. Hyper-parameter scales and the shrinkage weights are selected on the local folds (eval/tune.py), with a cross-fitted estimate for honesty. Selected scales: delta_var 1.00, tau2 0.21, beta_var 6.76, a_var 0.60, w_var 4.16. The coordinate search over the scales was run on a 40-subject-per-benchmark subsample, because a search under the full protocol costs about 20 s per candidate and does not fit the compute available here; the shrinkage weights are then fitted **and cross-fitted under the full protocol** (--scales ... --rounds 0 --crossfit). This is disclosed because the scales are therefore selected on a smaller sample than the one they are reported on.

4.3 Online update (submission/pirt_online.py)

One joint Gaussian over all latents of a benchmark. Each label is absorbed by an assumed-density-filtering step: the logistic likelihood is approximated by a probit, whose moments are closed-form, followed by a rank-one covariance update. Labels are processed one at a time in the order supplied; the state is cached and extended incrementally when the evaluator's label list grows. Labels of other subjects on the same benchmark update the shared latents (β, a, d_i, w_f). A numpy-free scalar fallback runs if numpy is unavailable.

4.4 Prediction and safety net

$p = E[\sigma(\text{logit})]$ under the posterior (probit approximation), then $p' = (1 - w_n) \cdot p + w_n \cdot \bar{p}$ with $\bar{p}$ the training base rate and w_n chosen per label count on local validation. Selected weights: $w_0 = 0.535, w_1 = 0.141, w_3 = 0.000, w_7 = 0.015, w_{15} = 0.004, w_{31} = 0.003$; final $\bar{p} = 0.2816$.

w_n is the weighted least-squares optimum in which each prediction of a pair with n targets carries weight $1/n$, matching the way the metric averages; pooling the predictions instead would let the largest pairs choose w for every pair. On this corpus the two agree to within $1e-5$ of ALC, so the alignment is a correctness property rather than a source of gain.

$\bar{p}$ is the mean over training subject-benchmark pairs of each pair's accuracy, not the pooled response rate. This matters: pooling lets the largest benchmark set the target, which here would give 0.21 against a pair-weighted 0.31, and cost about 0.003 of B_0 .

4.5 Acquisition (labeling.py) — not shipped

Streaming decision per candidate. The value of a candidate is the expected reduction of the posterior variance of the pair ability $c = a \theta_s + v_s - \beta$ (Fisher information under the current posterior, including the candidate's difficulty uncertainty), relative to a generic unseen item: ρ . The query probability is $\min(1, q \cdot \rho^2)$ with $q = \text{labels_remaining} / \text{items_remaining}$; with $\rho \equiv 1$ it is exactly the evaluator's random policy.

A single pre-registered A/B (eval/ab.py, results/acquisition_ab.md) compared the random policy against Fisher information and against posterior uncertainty, with the decision rule fixed beforehand: ship only if ALC is lower **and** the 95 % interval of a paired bootstrap over benchmarks excludes 0.

policy	ALC	Δ ALC vs random	95 % CI (benchmark clusters)
random (official)	0.16875	—	—
Fisher information	0.16981	+0.00106	[−0.00071, +0.00345]
posterior uncertainty	0.16846	−0.00029	[−0.00321, +0.00448]

Neither interval excludes 0, so **the submission ships without labeling.py** and uses the official random policy. The A/B was first run on the 40-subject subsample (Fisher +0.00127, uncertainty +0.00156, both intervals straddling 0) and then re-run unchanged on the full protocol, which is the table above; the decision is the same under both, and the rule was not altered after seeing either. Posterior uncertainty turns marginally negative on the larger sample but its interval is four times wider than the effect. Fisher information is better at B_1 and B_{31} and worse at B_3 – B_{15} , where the ALC weight is concentrated; we report this as an observation, not as grounds for a third attempt.

5. Results (local hold-out)

Cross-validated, leave-one-benchmark-out, pair-macro Brier; lower is better; 147 pairs and 19 621 evaluation targets. The rows below are the fully out-of-fold model **before** hyperparameter tuning, so no selection of any kind enters them.

Predictor	ALC	B_0	B_1	B_3	B_7	B_{15}	B_{31}
Constant 0.5	0.25000	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500
Empirical mean (organizers)	0.23110	0.2500	0.3449	0.2310	0.1902	0.1774	0.1740
Online IRT	0.16875	0.2211	0.1932	0.1771	0.1593	0.1412	0.1248
Online IRT, own-pair labels only	0.19450	0.2211	0.2117	0.1985	0.1875	0.1777	0.1731

That is 27 % better than the organizers’ baseline on ALC. With the shrinkage weights fitted and cross-fitted under the same protocol, the **cross-fitted (honest) estimate** is **ALC 0.16515**, per-budget Brier 0.2061 / 0.1897 / 0.1724 / 0.1574 / 0.1408 / 0.1248.

Unseen model identities (27 of 147 pairs, calibration never saw the identity): Online IRT 0.18424, empirical mean 0.24960, constant 0.5 0.25000. Seen identities: 0.16526 / 0.22693 / 0.25000. The advantage does not depend on having met the model before.

Per fold the ALC is 0.1509, 0.1943, 0.2728 and 0.1860; the worst fold is swe_rebench, which contributes exactly one pair.

Calibration improves with labels: expected calibration error $0.122 \rightarrow 0.072 \rightarrow 0.058 \rightarrow 0.023 \rightarrow 0.027 \rightarrow 0.016$ across the six budgets (figures/calibration.pdf). At zero labels the predictor is visibly over-confident; this is what the shrinkage weight $w_0 = 0.535$ absorbs.

Figures: figures/learning_curves.pdf, figures/calibration.pdf; acquisition A/B: results/acquisition_ab.md; data survey: results/data_survey.md; code review notes: results/ocr_review*.md.

5b. Results (live, Codabench formative feedback)

The shipped artefact (dist/paiec_irt.zip, sha256 98555787...2cfe0f, 15,436 bytes) was submitted **five times unchanged**, three on 2026-10-01 (955485, 955728, 955767) and two on

2026-10-02 (957572, 957580). The rules state that “each formative evaluation uses a newly sampled subset of the hidden test data, capped at 1,000 unique subject-item pairs”, so the five runs differ only in which subject-benchmark pairs were drawn. We report them as a repeated measurement, because a single live score turns out to be far noisier than we expected, and the noise is the main thing we learned from them.

	955485	955728	955767	957572	957580	mean	SD	local	gap vs local
Brier	0.20624	0.20098	0.18124	0.18685	0.19461	0.19398	0.0102	0.17540	+0.0186 (1.8 σ)
ALC									
B ₀	0.30845	0.27314	0.25431	0.30584	0.28153	0.28465	0.0228	0.21634	+0.0683 (3.0 σ)
B ₁	0.25933	0.22918	0.22100	0.24012	0.21610	0.23315	0.0172	0.20171	+0.0314 (1.8 σ)
B ₃	0.19287	0.20820	0.17058	0.20076	0.19303	0.19309	0.0141	0.18436	+0.0087 (0.6 σ)
B ₇	0.17314	0.17904	0.15904	0.14204	0.17071	0.16479	0.0147	0.16570	−0.0009 (−0.1 σ)
B ₁₅	0.16742	0.17175	0.15398	0.13154	0.16892	0.15872	0.0167	0.15042	+0.0083 (0.5 σ)
B ₃₁	0.16840	0.16031	0.14893	0.13378	0.16701	0.15569	0.0145	0.13330	+0.0224 (1.6 σ)
ECE	0.16324	0.13985	0.13641	0.19109	0.14631	0.15538	0.0225	—	—

Each run drew 8 or 9 subject-benchmark pairs (8, 8, 8, 9, 9), and in every one the unweighted mean of the per-pair ALCs reproduces the reported total to six decimals, confirming that the official aggregation is an equal-weight mean over pairs. Run 3’s eight pairs share no subject with run 1’s. The sampled subjects — and, as runs 4 and 5 show, the number of them — are where the variance comes from: with eight or nine pairs per run, the identity of the draw dominates.

The measurement noise is the headline result. A single formative score has a standard deviation of about **0.010** on ALC, **0.014-0.023** per budget, and **0.022** on ECE. The spread across five identical submissions is 0.0250, 2.5 SD. This is larger than most of the differences one would want to act on. What the evidence does and does not support:

1. **The live result is worse than the local estimate, but not significantly so.** The mean gap is +0.019 on ALC, 1.8 SD. All five runs lie above the local estimate, so the direction is consistent across every draw; the magnitude is not established.
2. **Only the zero-label budget survives the noise.** B₀ is +0.068 above local at **3.0 SD**, consistent in all five runs, and it is the one per-budget reading that strengthened as runs were added (2.3 σ at n=3, 2.6 σ at n=4, 3.0 σ at n=5). B₁ did not: it was 1.7 σ at n=3 and is 1.8 σ now, with the fifth run’s 0.2161 nearly level with the local 0.2017. **The shortfall is at the zero-label prior specifically, not at “the low budgets” generally.**
3. **The middle budgets are indistinguishable, and we can show it rather than assert it.** B₇ read +0.0047 (worse than local) at n=3, flipped sign to −0.0024 at n=4, and sits at −0.0009 (−0.1 SD) at n=5. **One additional draw reversed the direction of a per-budget conclusion.** B₃ and B₁₅ are likewise within 0.6 SD. Any statement about an individual budget other than B₀ is reading the draw, not the method.
4. **At zero labels the method is at best level with predicting 0.5.** B₀ was above the constant-0.5 value of 0.2500 in all five runs (minimum 0.2543), but the mean margin is +0.035, only 1.5 SD, and the smallest is 0.2 SD. We report this as a consistent direction rather than a confirmed deficit.
5. **Leaderboard position is not a stable quantity here, but the gap to the leader is.** The five scores span 0.1812–0.2062, a range that covered roughly four places on the 21-entry public table at the time of measurement, so we do not quote a rank. Against the organisers’ own entry at 0.180113 our mean is +0.014 (1.4 SD) — worse in all five

runs but not significantly so, the closest run being only 0.0011 above it. Against the best entry at 0.117238 the gap is +0.077, **7.6 SD**: that one is established. The 27 % margin over the empirical-mean baseline in §5 is a statement about that baseline method on local hold-out data and should not be read as competitiveness.

Two further remarks on procedure. First, the noise calibration should have been run before any inference was drawn from a live score, not after. Our first live number was used to diagnose a deficit, propose three fixes and revise protocol conclusions, all before we knew its standard deviation was 0.010 — with hindsight that first “finding” was under 3 SD and should have been held. Second, $\hat{\sigma}$ itself is estimated from five points and carries roughly 35 % relative error, so it sits close to the 0.01 threshold we pre-registered for discounting single-score inferences. We have not spent further submissions trying to resolve which side of that threshold it falls on: at this magnitude the noise is of the same order as every effect we would act on, and the third decimal of $\hat{\sigma}$ does not change that.

One of the five runs (957572) was later marked **Failed** by the platform, having already produced a complete scoring output whose nine per-pair ALCs average to the reported 0.186854 to six decimals. We keep it: the measurement exists and was checked, and the platform’s terminal states on this competition are demonstrably unreliable — of the ten submissions of this artefact, two failed outright, one has been stuck in Submitting for over two days and two more produced nothing in eleven hours. Dropping it is nevertheless worth reporting, because of what it does. The headline is unmoved: $\sigma(\text{ALC})$ goes from 0.0102 to 0.0108, the ALC gap from 1.8 to 1.9 SD, B_0 from 3.0 to 2.8 SD — both statements this report rests on survive either way. But B_{31} moves from 1.6 SD to **3.1 SD**, B_{15} from 0.5 to 1.9, and B_7 from -0.1 to $+0.6$. **Removing one draw out of five turns one per-budget reading from “indistinguishable” into a three-sigma effect.** That is the clearest statement of this section we can make: at this sample size the per-budget numbers are a property of the draw, and only B_0 and the total are worth quoting.

We pre-registered and tested three mechanism-level fixes for the zero-label prior: shrinking towards 0.5, estimating the target benchmark’s success rate online with a 0.5 cold start, and the same with the original constant as the cold start. Gates were fixed before running; **all three failed** on leave-one-benchmark-out, each making B_0 and/or B_1 worse. The constant was fitted on exactly the four public benchmarks, so for them it is close to the best available prior rather than a wrong one. The live shortfall at B_0 — now the one per-budget effect that clears the noise — appears to involve a benchmark whose success rate lies outside the 0.148–0.475 range the public pool spans, which no local protocol can reproduce, so we record the hypothesis as **untested rather than refuted**. The shipped configuration is unchanged.

No model choice in this report was made using live feedback. Formative evaluation re-samples on every submission and the summative evaluation uses a separate common subset, so selecting on it would be selecting on noise. All five runs above were spent on measuring that noise, not on choosing anything. Of the ten submissions of this artefact, two failed outright on the platform and three have remained queued without producing output (one for over 33 hours); we report the five that scored.

6. Limitations

- **The training pool is very small and narrow.** Four benchmarks, all software engineering / agent tasks; no mathematics, no document understanding. The private test benchmarks may well include domains with no representative here, and the across-benchmark priors that the method depends on are estimated from four points.
- **The local estimate is coarse.** Leave-one-benchmark-out over four benchmarks gives per-fold ALC of 0.151, 0.194, 0.273 and 0.186. Any difference smaller than that spread — including the 0.004 gained by hyper-parameter tuning — is not resolvable with this data. swe_rebench contributes a single subject-benchmark pair, and on that fold the model

(0.273) is worse than a constant 0.5; a benchmark with one subject gives the offline calibration nothing to learn from.

- **The live estimate is coarse too, and for a different reason.** Each formative evaluation draws eight or nine subject-benchmark pairs afresh, so a single live score carries a standard deviation of about 0.010 on ALC and 0.014–0.023 per budget (\$5b, five identical submissions). That is the same order as every effect in this report we might want to act on, so the live feedback supports only three statements: the gap to the leading entry is real (7.6 SD), the zero-label shortfall against our local estimate is real (3.0 SD), and nothing else is resolved — including whether we are behind the organizers’ own entry (1.4 SD). We measured this only after drawing conclusions from the first live score, which was a mistake in sequencing, not just in arithmetic.
- **The pre-registered bootstrap is weak by construction.** Resampling four benchmark clusters yields a coarse distribution. We kept the rule rather than changing it after seeing the data, and report a pair-clustered interval alongside as a diagnostic; both agree that the acquisition variants are not distinguishable from random.
- **Zero-label prediction is the weakest budget, but it does beat a constant.** B_0 is 0.2061 cross-fitted. The right comparison is a constant chosen the same way the model is fitted — the base rate of the training folds only — which scores 0.2122; the model is 0.006 better. An earlier draft of this report claimed the opposite by comparing against the constant that minimises error on the evaluation pool itself (0.2160 at the smaller protocol), which is not a baseline any predictor could have chosen in advance. This is a statement about local hold-out data against a base-rate constant, and it does not carry over to the live setting: \$5b finds live B_0 above the constant-0.5 value of 0.2500 in all five runs, and above our own local B_0 by 3.0 SD. The zero-label prior is the one place where local and live disagree beyond the noise. The weighted least squares for w_n described in §4.4 was aligned with the metric after that finding and changed the result by less than $1e-5$ of ALC, confirming that the earlier gap was an artefact of the unfair baseline, not a weighting bug.
- **reasoning_effort has no support.** The configuration-effect machinery for it is inert on this corpus; if the private subjects carry that field, its effect is untrained.
- Items of the evaluation pool are never labeled, so item-level variation comes only from visible features; the text of items is not used by the model beyond feature tokens. The text-difficulty prior (Task 4 of our plan) was not built.
- One latent ability dimension plus benchmark-specific deviations; skills that do not correlate with general ability are captured only through v_s .
- ADF is order-dependent and approximates the posterior.
- Evaluation worker concurrency is unstated in the rules; the engine is thread-safe but has not been tested under the organizers’ actual harness.

7. Reproducibility

```
python3.11 -m venv .venv && . .venv/bin/activate
pip install -r requirements-lock.txt
git clone --depth 1 https://github.com/aims-foundations/paiec_baseline # pinned at 82d330d
export PAIEC_BASELINE_DIR=<path to paiec_baseline>
export HF_TOKEN=<read token; the dataset is gated>
CAPS="--max-subjects 200 --max-eval 200 --max-stream 800"
python scripts/download_measurement_db.py --out data/raw
python scripts/survey_data.py --data data/raw --out results/data_survey.md
python -m eval.run_cv --data data/raw --variants const0.5 empirical_mean irt irt_pairscope \
    $CAPS --out results/by_budget.csv --records results/cv_records.npz
python -m eval.tune --data data/raw --crossfit --out results/tuned_hyper.json # scale search, 4
subject subsample
python -m eval.tune --data data/raw --crossfit --rounds 0 --scales results/tuned_hyper.json \
```

```

$CAPS --out results/tuned_hyper_full.json # shrinkage under the full p
python -m eval.ab --data data/raw $CAPS --out results/acquisition_ab.md
python tools/make_figures.py --records results/cv_records.npz --curves irt empirical_mean irt_pai
--calibrate irt --ab "irt:random" "irt+uncertainty:Posterior uncertainty" --ab-
records results/ab_records.npz
python -m train.train_final --data data/raw --tuned results/tuned_hyper_full.json
python tools/build_submission_zip.py
python $PAIEC_BASELINE_DIR/check_submission_zip.py dist/paiec_irt.zip

```

Seeds are fixed (split seed 0, deterministic SHA-256 hashing). Verified on Python 3.11.16 with the locked dependency versions: two `run_cv` runs produce identical CSVs, two `train_final` runs produce identical `params.json`, and two ZIP builds are byte-identical to each other. The organizers' `check_submission_zip.py` passes on the shipped artefact without `--static-only` (re-checked before release).

The rebuilt ZIP does not have the same hash as the scored one, and this is expected.

The artefact that was scored is sha256 98555787478009fc9e53b05a7a532912db37b716e24ed9abac121bac33c2f running `tools/build_submission_zip.py` against the source as published gives sha256 21a9df6f4323d94828280e8d6514c0579bb92b0f87b2ddc52371079279743a8a. Four of the five packaged files are byte-identical; the difference is confined to `pirt_online.py`, which gained an inert `shrink_target == "bench"` branch while the three zero-label candidates of \$5b were being tested after the artefact was built. The shipped `params.json` carries no target key, so the default "const" path runs and the two versions are not merely equivalent in principle: over 300 randomized (subject, item, label-sequence) triples with 0 to 31 labels they return bit-identical predictions, with a maximum absolute difference of 0. We kept the falsified candidates in the source rather than stripping them, because a candidate that was pre-registered and refuted is a result, and deleting it would make the record of \$5b unverifiable. Extracted into an empty directory and imported in a fresh process, the model predicts in single-digit microseconds per call (6.7 μ s with numpy, 2.8 μ s on the numpy-free fallback), which is negligible against the 8-hour run limit; malformed labels are skipped and an empty label list returns the prior.

The submission ZIP contains `model.py`, `pirt_online.py`, `params.json`, `requirements.txt` and `README.md`. No `models.txt` is needed: the submission declares no Hugging Face model. The build script scans every packaged file for credentials and refuses parameters fitted on synthetic data.

8. Disclosures

- **Pretrained models:** none. The submission contains no neural network, embedding model or LLM, and makes no API calls. Nothing in the evaluation path touches the network.
- **Data:** only the public (access-gated) `aims-foundations/measurement-db` at revision `bc8204d811823da849c6686bf124d4ca9f82e4de`; see §2.
- **LLM coding assistant:** Claude Opus 5 (Anthropic), used through the Claude desktop app's agentic coding mode, to write the code and draft this report under the author's direction; all results were produced by the scripts in this repository.